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Therapeutic-oligonucleotides activated by nucleases (TOUCAN): A nanocarrier system for the specific delivery of clinical nucleoside analogues

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ABSTRACT

Nucleoside analogues have been in clinical use since 1960s and they are still used as the first therapeutic option for several cancers and viral infections, due to their high therapeutic efficacy. However, their wide clinical acceptance has been limited due to their high toxicity and severe side effects to patients. Herein, we report on a nanocarrier system that delivers nucleosides analogues in a target-specific manner, making nucleoside-based therapeutics safer and with the possibility to be used in other human conditions. This system, named, Therapeutic OligonUCleotides Activated by Nucleases" (TOUCAN) combines: i) the recognition power of oligonucleotides as substrates, ii) the use of nucleases as enzymatic biomarkers and iii) the clinical efficacy of nucleoside analogues, in a single approach. As a proof-of-concept, we report on a TOUCAN that is activated by a specific nuclease produced by bacteria and releases a therapeutic nucleoside, floxuridine. We demonstrate, for the first time, that, by incorporating a therapeutic nucleoside analogue into oligonucleotide probes, we can specifically inhibit bacterial growth in cultures. In this study, *Staphylococcus aureus* was selected as the targeted bacteria and the TOUCAN strategy successfully inhibited its growth with minimal inhibitory concentration (MIC) values ranging from 0.62 to 40 mg/L across all tested strains. Moreover, our results indicate that the intravenous administration of TOUCANs at a dose of 20 mg/kg over a 24-h period is a highly effective method for treating bacterial infections in a mouse model of pyomyositis. Importantly, no signs of toxicity were observed in our *in vitro* and *in vivo* studies. This work can significantly impact the current management of bacterial infections, laying the grounds for the development of a different class of antibiotics. Furthermore, it can provide a safer delivery platform for clinical nucleoside therapeutics in any human conditions, such as cancer and viral infection, where specific nuclease activity has been reported.

1. Introduction

Nucleoside analogues (NA) are synthetic compounds with similar chemical structure to the natural nucleosides (DNA and RNA), that mimic their counterparts in terms of uptake and metabolism. NA enter cells *via* specific nucleoside transporters [1]. Once inside the cell, they are phosphorylated by viral or cellular kinases to the active 5-triphosphate derivatives. NA exert their action by incorporation into the newly synthesized DNA, resulting in synthesis inhibition, chain termination and cell death [2,3]. Both, bacteria and eukaryotic cells can uptake nucleoside analogues passively, by diffusion, or actively, through transporters in their cell membranes [4,5]. The main route of uptake is the active transport, which allows these compounds to accumulate at higher levels inside the cell, especially when passive diffusion is not effective [5]. NA are among the most important class of antimetabolite drugs in the clinical setting and are used widely as both, anticancer and antiviral agents [6,7]. However, the high toxicity and strong side effects associated to NA have limited their clinical use [8]. Despite their

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debilitating side effects, systemic administration of naked NA is still the main formulation used in the clinic for certain diseases, like cancer [9], however their broader use and their utility in other human pathologies is very limited. One clear example of the applicability limitation of NA is their use in bacterial infections. Despite the significant number of reports indicating the utility of NA to treat bacterial infections [10], along with the use of NA derivatives that are clinically approved, there is not a single NA implemented in the clinic as antibiotic. Two plausible factors could be responsible for the poor use of NA as antibiotics in the clinic: i) the high toxicity associated with the treatment and ii) the availability of other antibiotic alternatives. However, the emergence of multi-drug resistant pathogens, and the possibility to re-purpose the NA already approved for clinical use, has opened the possibility of using these drugs as antibiotics [10]. Thus, technologies that can safely deliver NA to target cells could fully unlock the therapeutic potential of NA. Herein, we report on Therapeutic OligonUCleotides Activated by Nucleases (TOUCAN) as a novel therapeutic strategy that delivers NA in a safer manner. TOUCANs combine the power of specific target recognition of oligonucleotides [11], the utility of nucleases as biomarkers and the clinical efficacy of NA [10] in a single approach, taking advantage of their best performing properties to provide a strategy with great potential for clinical translation. The main idea of this approach emerged from our previous findings about the utility of nucleases as biomarkers of disease [11–13]. Nucleases are enzymes that catalyze DNA and RNA cleavage with remarkable efficiency and are indispensable for life. Nucleases are highly diverse and their evolution over millions of years offers the possibility of identifying specific blueprints of prokaryotic and eukaryotic cells [14,15]. Moreover, nucleases are known to be virulence factors of pathogenic bacteria, trait that we have been exploiting in our lab as biomarker for detection of several bacteria [16], such as *Staphylococcus aureus* [11]. In addition, the availability and use of chemically modified nucleosides, such as 2'-OMe, 2'-F, LNAs, and many others, enhance the stability of oligonucleotides to nuclease degradation, increase binding affinity and add other unique properties that offer the possibility of tailoring the oligonucleotide substrates for specific identification of target nucleases [17]. Thus, the high diversity and specificity of nucleases as biomarkers and the use of oligonucleotides as substrates provide the framework for developing novel diagnostic [11,13,18,19] and therapeutic approaches, such as TOUCANs. In a previous study, we have identified a specific oligonucleotide probe, TTprobe, that targets micrococcal nuclease (MN), a nuclease derived from *S. aureus*. This TTprobe is a short oligonucleotide (FAM-mC-mU-mC-mG-T-T-mCmGmUmUmC-RQ) consisting of a specific moiety -TT- (DNA thymine) as substrate for MN and flanked by a resistant moiety consisting of chemically modified 2'-*O*-methyl nucleosides (-mC-, -mU- or -mG-), to avoid non-specific degradation. The TTprobe was modified at the ends with a fluorophore (FAM) and quencher (RQ) to provide detection capabilities [11]. Building on this knowledge, we hypothesized that, by incorporating a nucleoside analogue into the TTprobe, we can create a novel oligonucleotide therapeutic strategy (TOUCAN) for treating *S. aureus* infections in a safer and more efficient manner. To demonstrate the feasibility of TOUCAN technology, we have created the first-in-class antibiotic, based on short sequence oligonucleotides programmed as a smart nanocarrier system, with 7 nucleotide long (7mer) of approximately 2.38 nm in size (7mer x 0.34 nm per nucleoside). As far as we know, this 7mer approach is the smallest therapeutic oligonucleotide reported up to date.

It is noteworthy that for systemic administration, particularly intravenous administration, therapeutic oligonucleotides require chemical modifications [20,21]. Among various available modifications, phosphorothioate backbone modifications are widely utilized for delivering naked or unformulated oligonucleotides (*e.g.* liposome/capsule free) [22–24].

These modifications lead to a significant increase in plasma protein binding, exceeding 90%, a half-life of approximately 2 h, and rapid transfer from the bloodstream to diverse tissues such as the kidney, liver, and spleen. Additionally, they display limited accumulation in skeletal muscle [25–28]. In contrast, there is an extensive body of evidence indicating that phosphate backbone oligonucleotides (unmodified) are rapidly cleared from plasma and excreted at higher levels in urine during the initial minutes and hours after administration [29,30]. This results in reduced cell internalization and tissue accumulation, rendering such oligonucleotides inefficient for cellular uptake [29–36].

Considering these facts, TOUCAN oligonucleotides were designed with a phosphate backbone (unmodified) to minimize the oligonucleotide cell uptake and enable specific nuclease degradation. This design renders TOUCAN into a delivery system that becomes active solely upon encountering the target nuclease. Subsequently, this leads to the release of the nucleoside analogue that is internalized only in its monomeric form, through membrane-bound nucleoside transporters, rather than the oligonucleotide form.

TOUCAN consist of three moieties, each having a different function. The resistant moiety (**Fig. S1**, black) confers protection against degradation from non-specific nucleases, while the specific moiety (**Fig. S1**, yellow) allows the targeting of the infection-causing bacteria. The drug moiety (**Fig. S1**, green) is the nucleoside analogue that, in this proof-of-concept study, works as an antibiotic. Fig. 1 shows the TOUCAN's mechanism of action in a mouse pyomyositis model. Briefly, once TOUCANs reach the infection site (bacteria in orange), they are specifically degraded by the *S. aureus* nucleases (MN) releasing the drug cargo moiety. Subsequently, the drug is internalized by the bacteria, resulting in cell death (bacteria in brown). Hence, this pro-drug mechanism can offer a safer, simpler, and more attractive strategy to treat infections (See animation). Importantly, TOUCAN concept allows the repurposing of a long list of clinically approved nucleoside analogues [10], that are currently underused due to high toxicity. We envision the great potential of this technology to be translated to other human conditions such as cancer, where nucleases have been reported as potential biomarkers [37].

2. Results and discussion

2.1. Incorporation of a therapeutic nucleoside into oligonucleotide TTprobe

We sought to demonstrate that the incorporation of a nucleic acid analogue into the TTprobe [11] preserves its recognition properties for *S. aureus*. Therefore, we chose to incorporate in our strategy a nucleoside antimetabolite, floxuridine (Fx), which is often used for the chemotherapeutic treatment of colorectal cancer [38] and also indicated for renal carcinoma [39] and metastatic liver disease [40]. Several studies have also demonstrated the antibacterial potential of Fx by reporting efficient growth inhibition of *S. aureus* at low minimal inhibitory concentrations (MICs) [10,41–45]. Furthermore, to facilitate the incorporation of Fx into the TTprobe, we have synthesized Fx in the phosphoramidite format as described in **Fig. S2** and the methods section. In this study, we have engineered our previously reported TTprobe [11] into a simpler sequence with only two types of nucleosides (5'-UUUU**TT**UUUU-3′), named TTprobe2, where the U stands for 2'-*O*-methyl uridine; and **T** stands for deoxythymidine. To demonstrate that TTprobe2 parallels the efficiency of the original TTprobe, the nuclease activity assay was performed using cultures of *S. aureus* (MN+) and control cultures of *Staphylococcus epidermidis* (*S. epidermidis*), bacteria that does not produce MN (MN-). Our results confirmed the efficient degradation of the TTprobe2 by *S. aureus*, while the probe was rendered stable in the presence of *S. epidermidis* (Fig. 2a). Furthermore, we sought to investigate whether the incorporation of therapeutic nucleoside analogues, such as Fx, into the TTprobe2 can provide therapeutic efficacy against *S. aureus*. Hence, Fx nucleoside was incorporated into the TOUCAN structure using two strategies: i) upstream FxTT (5'-UUUU**FxTT**UUUU-3′), and ii) downstream TTFx (5'UUUU**TTFx**UUUU-3′). We have also included a Fx-control sequence without the TT motif

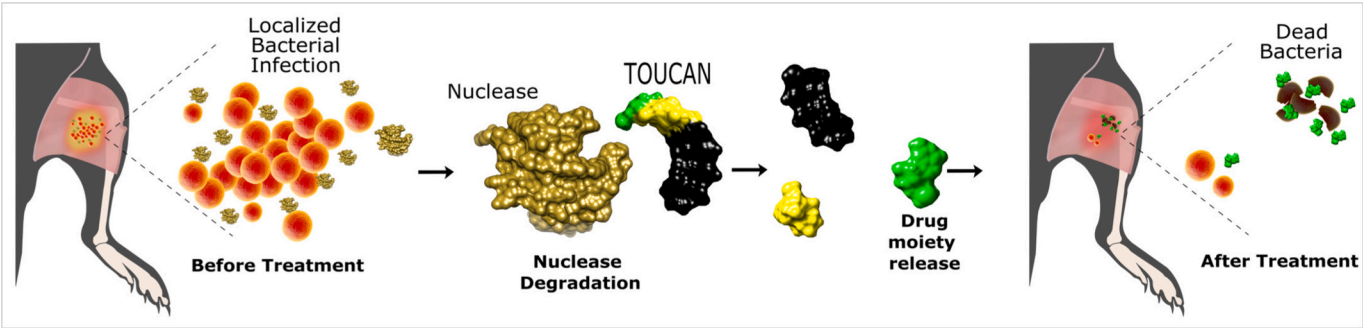

Fig. 1. TOUCAN technology. Schematic representation of TOUCANs mechanism of action in a localized *S. aureus* infection, in a mouse pyomyositis model.

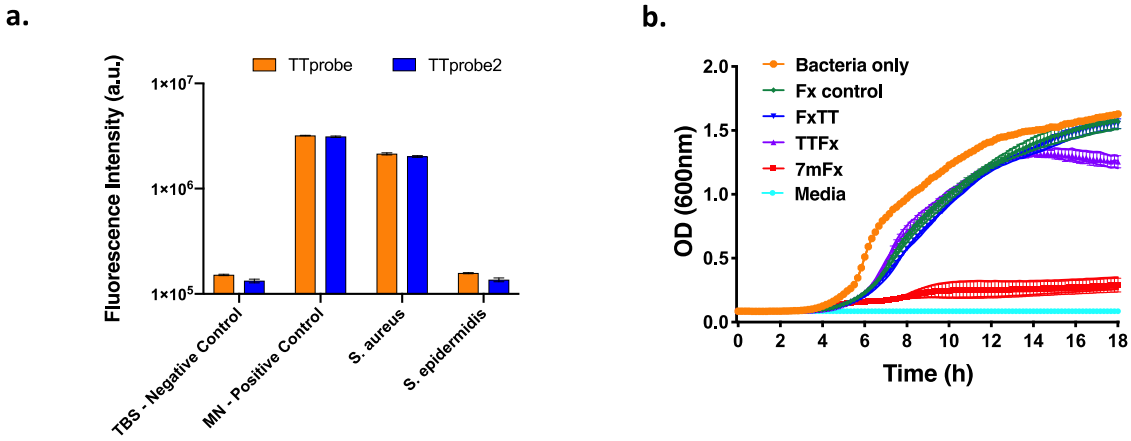

Fig. 2. Feasibility of TOUCAN using *in vitro models*. a) Validation of TTprobe2 as a simpler oligonucleotide probe for TOUCAN development. b) Growth inhibition curves of TOUCAN with the nucleoside analogue (Floxuridine, Fx) incorporated at different positions into the TTprobe2. Each data point represents the mean ± SD, $n = 3$.

(5'-UUUU**Fx**UUUU-3′), which is expected to be non-degradable. The efficacy of the specific release of Fx was demonstrated in *S. aureus* cultures by measuring the optical density (OD) over 18 h. TOUCANs Fx-control and FxTT have shown very low growth inhibition efficiency in *S. aureus* cultures (Fig. 2b, green and blue lines, respectively), whereas TTFx inhibited 24.2% of bacterial growth, (Fig. 2b, purple line) indicating that downstream incorporation (TTFx) facilitates the therapeutic activity of Fx. Based on these results, we designed an additional TOUCAN consisting of the first 7mer of the TTFx oligonucleotide, that we named 7mFx (5'-UUUU**TTFx**-3′). This design is based on the rationale that the TOUCAN will be cleaved at the TT site, thus releasing the Fx as therapeutic cargo. The 7mFx design is aligned with the therapeutic mechanism of action of Fx that involves: i) crossing the bacterial membrane, ii) intracellular phosphorylation and iii) inhibition of DNA synthesis [46–48]. We thus expect to see an increase in the TOUCAN 7mFx's efficacy. TOUCAN 7mFx was synthetized and evaluated under the same conditions as previously described for the other TOUCANs. Fig. 2b (red line) shows high therapeutic efficacy of TOUCAN 7mFx (40

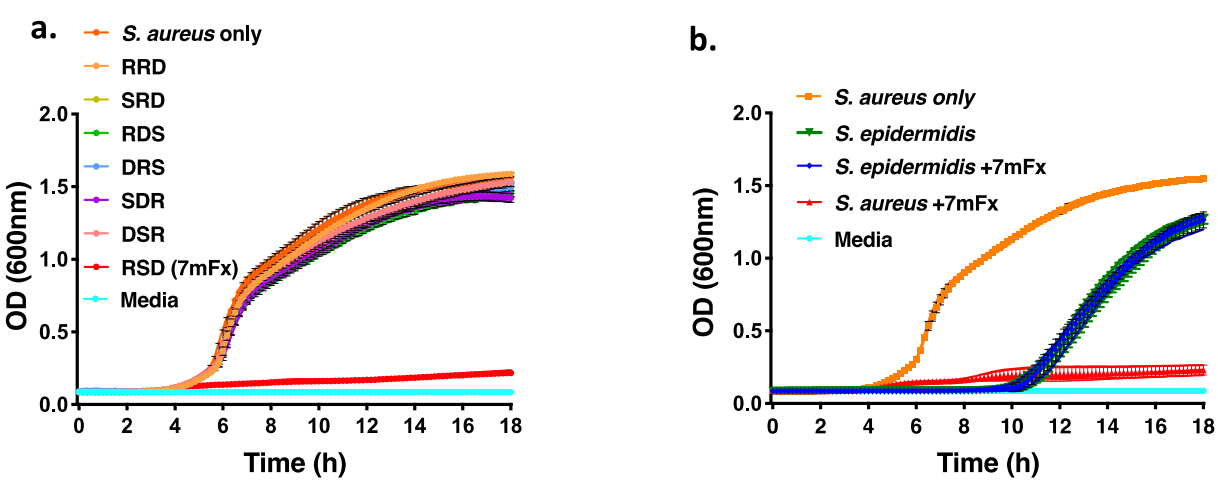

Fig. 3. Modularity and specificity of TOUCANs. a) Growth curves with all combinations of the three TOUCAN moieties; Resistant (R), Specific (S) and Drug (D). b) Growth curves of TOUCAN for targeted (*S. aureus*) and non-targeted (*S. epidermidis*) bacteria. Each data point represents the mean ± SD, n = 3.

mg/L) with the capability of inhibiting 86.8% of the *S. aureus* growth.

2.2. Modular and functional features of TOUCAN technology

We interrogated whether the location of each moiety plays a role in the functionality of TOUCANs. We hypothesized that the order of these three moieties (R, resistant moiety; S, specific recognition moiety; and D, drug moiety) within the structure of TOUCAN is critical for their therapeutic efficacy. To validate our hypothesis, we have synthesized the TOUCAN 7mFx with all possible combinations of the R-S-D moieties, along with the RRD-probe control (without the TT motif). Fig. 3a (red line) reports the therapeutic efficacy of all combinations of the three moieties, clearly indicating that RSD (7mFx at 40 mg/L) is the only combination exhibiting high therapeutic efficacy. **Table S1** provides a detailed sequence description of all combinations analyzed and all sequences used in this study. These results confirm that TOUCAN system is a modular approach comprised of three moieties that need to be arranged in a specific order to provide high therapeutic efficacy.

2.3. Specificity and safety studies in S. aureus cultures

Next, we sought to evaluate the safety and specificity of the lead TOUCAN by treating cultures of *S. aureus* (target) and *S. epidermidis* (control) with 40 mg/L of 7mFx (RSD). The growth curves in Fig. 3b show high growth inhibition of *S. aureus* (red line) by TOUCAN 7mFx, with minimal associated toxicity to *S. epidermidis* (blue line), confirming the safety profile for non-targeted bacteria. These results are consistent with all reported growth inhibition curves in this study and confirm the feasibility and specificity of TOUCANs. It has been previously reported that the antibiotic response of different strains within the same species can differ based on several mechanisms. Those mechanisms involve metabolic and growth rates, drug resistance, antimicrobial receptors [49] and more importantly, in the case of TOUCANs, the functionality of nucleoside transporters [50] and kinases [51]. We performed specificity studies, by testing Fx (drug only) and TOUCAN 7mFx against nine common strains of *S. aureus* (ATCC collection); from which eight strains are methicillin-resistant *Staphylococcus aureus* (MRSA), and one is methicillin-susceptible *Staphylococcus aureus* (MSSA), along with six non-targeted bacteria used as controls. **Table S2** shows the MIC studies for Fx (drug only) and TOUCAN 7mFx that were carried out under the same conditions as described in methods section. Both, drug only and TOUCAN were able to inhibit the growth of all *S. aureus* strains tested with MIC values from 0.005 to 0.32 mg/L for Fx, and TOUCAN 7mFx has reported MIC values from 0.62 to 40 mg/L (**Table S2**). Notably, efficient growth inhibition was observed for all MRSA strains tested with Fx and TOUCAN 7mFx.

2.4. Variability of nuclease activity in S. aureus strains

To discard the possibility that the MICs variability observed among strains was caused by low amounts of nuclease production by some strains, we have performed nuclease activity assays for each strain. TTprobe2 was modified with a fluorophore and a quencher, as previously described [11], allowing for the quantification of the nuclease activity. **Fig. S3** shows that all *S. aureus* strains can degrade TTprobe2 efficiently, with similar fluorescence signals, suggesting that nuclease activity is not the cause for variability among strains.

2.5. Dose-response studies

Among all the *S. aureus* strains tested, the treatment with TOUCAN 7mFx was most efficient on the ATCC1766 (MRSA) strain. In order to identify the best model for further animal experimentation, we carried out TOUCAN's dose response experiments with this MRSA strain, along with *S. epidermidis* cultures as control bacteria (Fig. 4). TOUCAN 7mFx has shown remarkable 99.4% bacterial growth inhibition at concentrations from 0.62 to 10 mg/L, measured by OD and visual inspection (Fig. 4a – 4b, respectively). Also, partial inhibition could still be seen at the lower concentration of 0.31 mg/L. In contrast, *S. epidermidis* cultures, used as control, were not affected by addition of 7mFx at all concentrations tested (Fig. 4c and d), underscoring the high specificity of the TOUCAN 7mFx. The raw data of dose response curves and the plate inspected by naked eye after 18 h are shown in **Fig. S4**. These results suggest the high inhibitory effect on *S. aureus* growth achieved by TOUCAN 7mFx.

2.6. Laboratory evolution of resistance for S. aureus under the exposure of TOUCAN 7mFx and floxuridine

As an initial attempt to assess the development of resistance to the TOUCAN approach, we conducted laboratory evolution tests on *S. aureus* using both, floxuridine and TOUCAN 7mFx, as previously described [52]. Our aim was to observe the evolution of antimicrobial resistance by exposing *S. aureus* to suboptimal doses of both compounds, specifically 1/2 MIC, for six consecutive days and analysing the fold MIC shift for each compound after each exposure. The results are depicted in Fig. S5 and S6, which illustrate the phenotypical shift in MIC for floxuridine and TOUCAN 7mFx, respectively. Next, we compared the fold MIC shift for both compounds, as shown in Fig. S7, and found that TOUCAN 7mFx did not develop resistance during the 6 days of exposure,

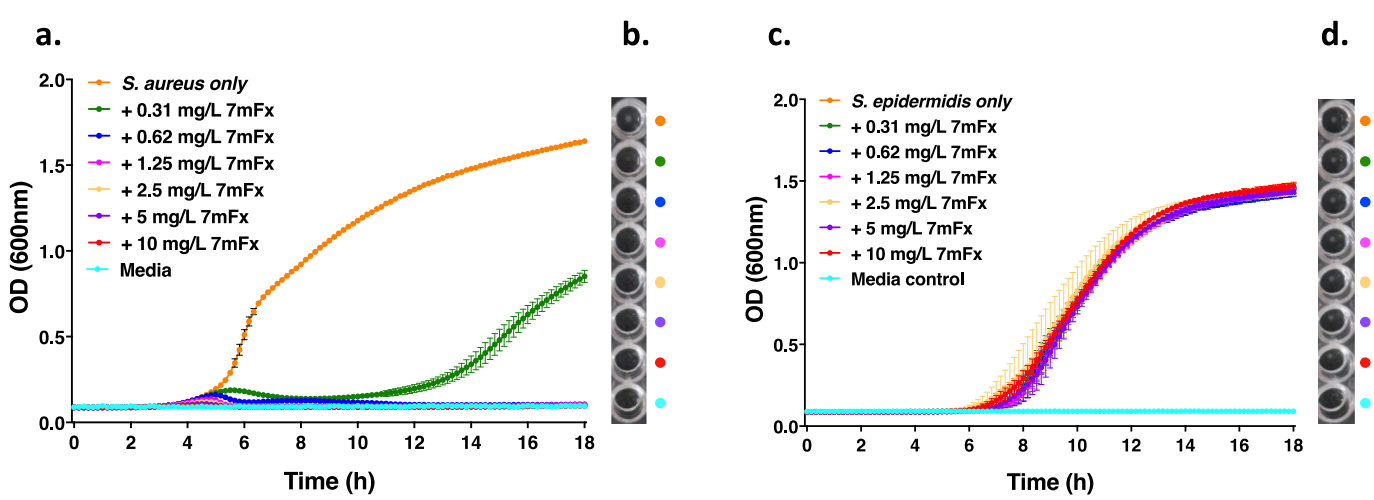

Fig. 4. TOUCANs dose response curves for *S. aureus*. TOUCAN 7mFx dose response curves and MIC for a) targeted bacteria (*S. aureus*) along with c) control bacteria (*S. epidermidis*) evaluated by continuous monitoring of OD at 600 nm and by naked eye after 18 h (b and d, respectively). Each data point is the mean ± SD, n = 3.

with only a 2-fold difference observed on day 6. In contrast, floxuridine displayed a significant increase in fold MIC shift from 8-fold on day 2 to 32-fold on day 6. These findings suggest that the TOUCAN approach is less prone to develop antimicrobial resistance. However, these results are preliminary and require further confirmation through genotypical assessment and additional experimentation involving longer exposure times and varying concentrations of exposure.

2.7. Clinical validation of TOUCAN 7mFx

To validate the clinical potential of TOUCAN technology, efficacy studies with TOUCAN 7mFx were conducted, in parallel, by an independent laboratory with expertise in antimicrobial susceptibility testing. Thus, fifteen clinical bacterial specimens belonging to five high-risk clones of *S. aureus* [53,54] were isolated at Karolinska University Laboratory. Five sequence types (STs) of methicillin-resistant *S. aureus* (ST1 = US400, ST5 = US100/EMRSA-3, ST8 = US300/EMRSA-2, ST30 = US200/EMRSA-16, ST22 = EMRSA-15) were analyzed using TOUCAN 7mFx, along with cultures of *S. epidermidis* (ATCC 12228) as control. The broth microdilution method was performed under conditions described in the methods section, and the MICs values and clinical isolates information is reported in Table 1. Importantly, all tested *S. aureus* clinical isolates were susceptible to TOUCAN 7mFx with MIC values between 1.25 and 40 mg/L, results that were in good agreement with our previous *in vitro* characterization studies with other non-clinical strains (ATCC collection). These findings confirm the potential applicability of the TOUCAN technology in the clinical setting. Altogether, these results indicate TOUCAN 7mFx as being a promising candidate for *in vivo* validation, in a murine infection model.

2.8. In vivo validation of TOUCAN 7mFx in mouse thigh infection model

First, we analyzed the stability of TOUCAN 7mFx in mouse serum to anticipate any possible non-specific degradation by endogenous nucleases. The stability analysis of 7mFx was carried out by mass spectrometry (MALDI-TOF-MS) to observe the fragmentation profile with and without incubation with *S. aureus* nuclease (MN) as specific target, or mouse serum as non-specific target. **Fig. S8a** shows the mass spectra of 7mFx incubated in buffer (control), and after the treatment with MN at different time points, from 0 to 40 min (**Fig. S8b-g**). The results confirm that 7mFx was completely degraded by MN after 5 min and the fragmentation profile is consistent with the cleavage at the specific TT moiety. In contrast, limited degradation of 7mFx was observed after 20 min when incubated with mouse serum (**Fig. S9a-g**). These results suggest that limited non-specific cleavage can be expected in mouse serum, however most of TOUCAN 7mFx degradation occurs only in presence of *S. aureus* nucleases (MN). Furthermore, we optimized a mouse model of thigh muscle infection based on our previous work [11]. The therapeutic regime was adapted from a previous study which validated the efficacy of several antibiotics against *S. aureus* [55]. To begin, we conducted studies to ensure that mice could tolerate the proposed TOUCAN regimen of 20 mg/Kg/day administered in 4 doses every 6 h. Additionally, we tested a dose 5 times higher (100 mg/kg/day) as indicated in the methods section. To enable comparison, we evaluated the impact of administering only the drug floxuridine under the same conditions. Results from the tolerability tests demonstrate that the doses of TOUCAN 7mFx at either 20 or 100 mg/kg/day did not exhibit any indications of toxicity within 24 h. However, floxuridine at 20 mg/kg/day demonstrated tolerability, whereas at 100 mg/kg/day it exhibited signs of mild toxicity from 3 h onwards (Table S3). These results validate the safety profile of TOUCAN 7mFx at the therapeutic dose of 20 mg/kg/day. Subsequently, we carried out experiments to evaluate the therapeutic efficacy of TOUCANs. Briefly, mice were inoculated with bacteria intramuscularly, in one leg (right). The first dose of TOUCAN was intravenously administered at 2 h post-inoculation, followed by three additional injections of the same concentration every 6 h. Three groups of animals were evaluated: TOUCAN 7mFx, along with the control groups placebo and Fx (drug only). A detailed description of the animal model is provided in the methods section. Fig. 5 provides compelling evidence of the therapeutic efficacy of TOUCAN 7mFx for treating *S. aureus* infections in mice. *Ex vivo* mouse pictures show that placebo treated mice developed a visible bacterial infection (Fig. 5a, black arrow), and a slightly stronger infection was observed in mice treated with Fx (drug only) (Fig. 5c, black arrow). In contrast, no visible sign of bacterial infection was seen in the leg gross pictures of mice treated with TOUCAN 7mFx, suggesting that infection was successfully resolved (Fig. 5b). Importantly, Fx and TOUCAN 7mFx groups were treated with equivalent amounts of Fx. As expected, the uninfected legs (left) for all groups did not show evidence of bacterial infection. Replicates of gross pictures for the three groups are presented in **Fig. S10**. To confirm these findings, we carried out histopathological analysis and bacteria counts of the uninfected and infected leg tissues, along with the blood analysis. The histopathological analysis showed no evidence of bacterial infection or inflammation for the uninfected legs (Fig. 5d). In contrast, mice treated with placebo and Fx (drug only), Fig. 5e and Fig. 5g respectively, have shown clear evidence of bacterial infection, inflammation, and neutrophil infiltration (black arrow heads). On the other hand, minimal inflammation with residual neutrophil infiltration was seen in mice treated with TOUCAN 7mFx (Fig. 5f), suggesting that immunological clearance of the infection took place (black arrow head). **Fig. S11** shows the histopathological images acquired with different objectives (4×, 10× and 20×). Next, we counted the number of bacteria for each treated group (Fig. 5h). As expected, no detectable bacteria were found in the uninfected legs, while high bacterial counts were detected in the placebo and Fx groups. Interestingly, the number of bacteria recovered from the group treated with TOUCAN 7mFx was significantly lower (Fig. 5h), with 642 and 2949 times less bacteria when compared to placebo and Fx groups, respectively. Representative colony forming unit (CFU) count plates are provided in **Fig. S12**, where the difference in CFUs among groups can be observed. Furthermore, we analyzed the percentage of neutrophils in blood samples. Neutrophils are the first host responders to presence of pathogenic bacteria and therefore, they are very important for the evaluation of bacterial infections. For a thorough analysis, and to avoid biases, blood samples from an additional group of uninfected/untreated mice were also evaluated. The uninfected/untreated group has reported a basal neutrophil population of 9.8 ± 0.9%. In contrast, the blood from placebo and Fx treated mice have shown an important elevation in neutrophils count, up to 29.0 ± 11.9% and 36,1 ± 9.5% respectively. Mice treated with TOUCAN 7mFx have reported a slight increase in the neutrophils count (14,1 ± 1.9%) indicating that the

Table 1
Clinical validation of TOUCANs. External validation of TOUCAN 7mFx efficacy using clinical isolates of five high-risk clones of *S. aureus* isolated by Karolinska Institute (KI), Sweden. Five sequence types (ST) were analyzed and three clones per ST were tested. *S. epidermidis* was used as control. All cultures were evaluated by three replicates.

Isolate number	Sequence Type (ST)	MICs (mg/L)
KI-217	1	20
KI-469	1	2.5
KI-674	1	20
KI-117	5	5
KI-356	5	10
KI-921	5	20
KI-656	8	10
KI-728	8	20
KI-929	8	10
KI-588	22	40
KI-650	22	40
KI-742	22	5
KI-561	30	20
KI-831	30	5
KI-856	30	1.25
S.epidermidis ATCC-12228	Control	>40

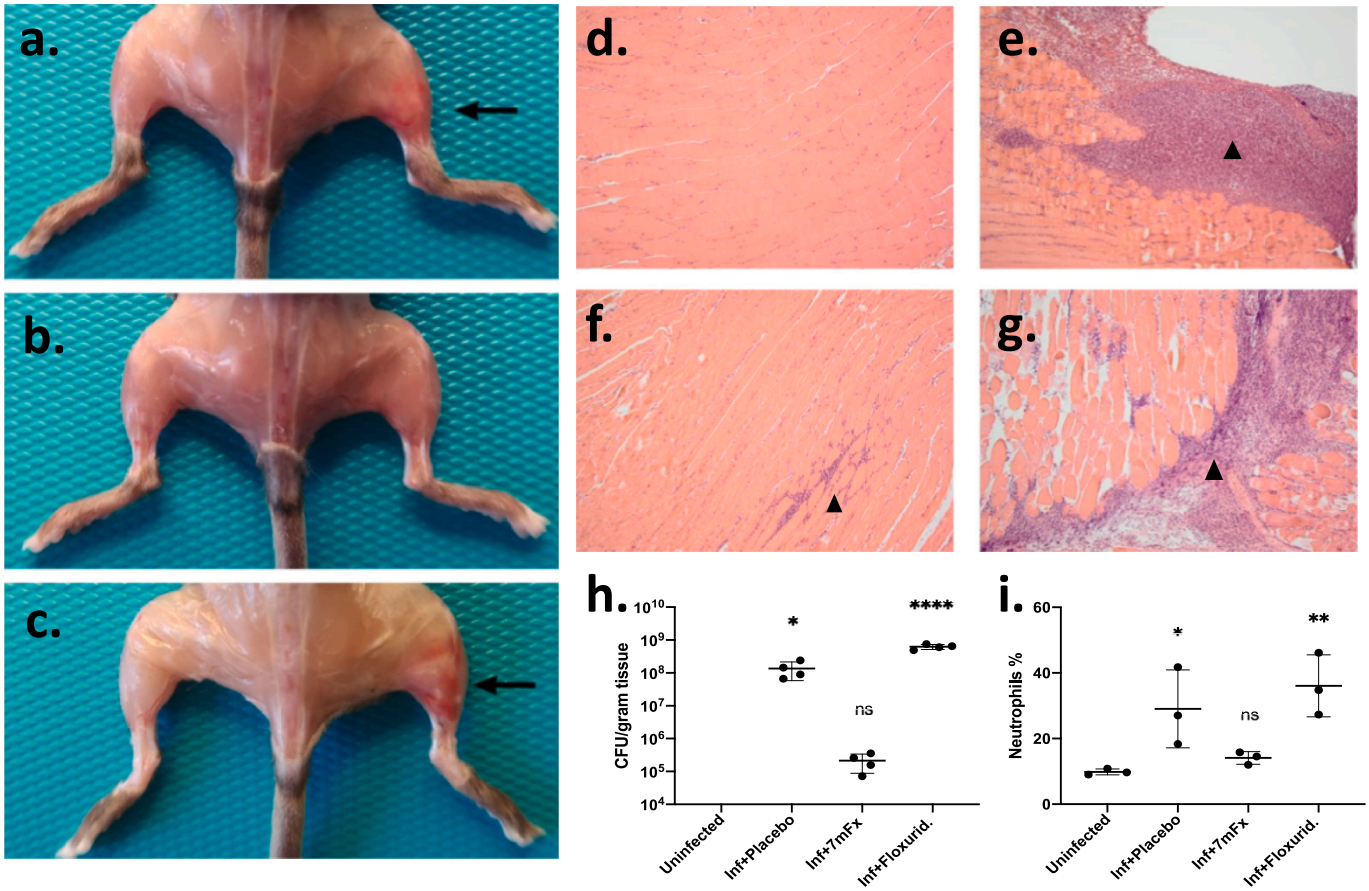

Fig. 5. Therapeutic efficacy of TOUCANs in mice. Representative gross photographs of mice groups treated with a) PBS, b) TOUCAN 7mFx and c) floxuridine. Histopathological evaluation of mouse leg tissues of d) uninfected/untreated, e) infected treated with placebo (PBS), f) infected treated with TOUCAN 7mFx and g) infected treated with floxuridine. All photographs are representative of at least 4 mice per group. h) Bacteria counts per gram tissue of uninfected and treated infected legs reported as CFU/g. i) Percentage blood neutrophils of uninfected and infected mice. Statistical analysis for the bacterial counts and neutrophils was performed with a one-way ANOVA using Dunnett's multiple comparison test. Data is presented as mean ± SD, $n = 4$ for CFUs and n = 3 for neutrophils. Statistical significance *vs.* uninfected leg control is denoted by **** $p < 0.0001$, * $p < 0.0332$ and ns =0.1234 stands for no significant difference when compared to uninfected leg.

bacterial infection was resolved (Fig. 5i). Complete blood counts are provided as supplementary information in **Table S4**.

Another interesting observation derived from the blood analysis was reduction of the red blood cells (RBCs) number and levels of hemoglobin in mice treated with Fx. However, no significant differences were observed for the uninfected, placebo and TOUCAN 7mFx treated groups (**Fig. S13**). This finding could explain the stronger infection observed in the Fx treated group as depicted by the gross pictures, along with the high CFU counts in the tissue and the highest population of neutrophils. RBCs reduction (anemia) is a marker of acute toxicity and suggests that, in part, the red blood cells were affected by the systemic administration of Fx (drug only). The absence of anemia in mice treated with TOUCAN 7mFx and the therapeutic efficacy observed in this group, suggest that most of the drug was released at the infection site with no indication of toxicity to RBCs. The results suggest that treatment with TOUCAN 7mFx at a dosage of 20 mg/kg/day has the potential to effectively and safely cure bacterial infections caused by *S. aureus*. Accurately translating the drug dosage from animal models to humans is crucial for the development of TOUCAN technology. While a therapeutic dose of 20 mg/kg/day of TOUCAN 7mFx was effective in mice, appropriate adjustments are needed to account for species-specific differences in drug metabolism and pharmacokinetics before determining the appropriate dose for humans. Moreover, a direct comparison between the dosage of TOUCAN 7mFx and floxuridine in clinical trials is not accurate due to the unique features of TOUCAN 7mFx. These features include its safety profile, targeted accumulation at the site of infection, efficient clearance, and reduced likelihood of resistance. Therefore, it is essential to conduct thorough preclinical and clinical studies to validate the therapeutic dose translation and ensure the safety and efficacy of TOUCANs in humans. Altogether, these results confirm that TOUCAN 7mFx can provide a targeted strategy to cure bacterial infections and demonstrate great potential as a first-in-class antibiotic.

3. Conclusion

NA are an important part of our arsenal against cancer and viral infections, and they represent one of the most relevant classes of antimetabolites in the clinical practice. However, the use of NA in other human pathologies, such as bacterial infections, is limited due to their high toxicity, and the balance of risks and benefits for the patient is still under debate. Safer systemic administration of NA could unlock their full potential for other clinical applications, such as the use as antibiotics, which is a clear unmet clinical need. Furthermore, in the last five decades, the development of NA incurred large investments in terms of money and time, with only few novel NA making it to the clinic. In this context, we believe that the TOUCAN technology can provide a platform to revive any potent NA that has failed clinical trials or has been withdrawn from the market due to their strong side effects and significant toxicity. In this study, we demonstrate the utility of TOUCANs as a novel therapeutic strategy based on oligonucleotides. Our *in vitro* and *in vivo* results confirm that TOUCAN 7mFx can provide a targeted strategy to cure bacterial infections and demonstrate great potential as a first-in-

class targeted antibiotic. Importantly, the lead TOUCAN 7mFx was clinically validated by an independent laboratory, confirming the applicability of this technology in the clinical practice. We have also shown that TOUCANs feature a modular framework that allows incorporation of different drug moieties, thus offering the possibility of formulating TOUCANs with different drugs that could provide a synergistic effect. We believe that TOUCAN technology can help treat difficult infections caused by multidrug-resistant bacteria by repurposing NA as antibiotics, while decreasing the probability of developing antimicrobial resistance in the formulated drug. Although promising technologies such as bacteriophages are striving to overcome significant limitations with intracellular delivery, spectrum of activity and off-target effects, herein we show that TOUCAN technology could be a step closer to the clinical practice, initially for targeting bacterial infections. Moreover, the possibility of using TOUCAN technology in other human diseases such as cancer, viral infections, or autoimmune disorders is feasible, since specific nucleases have been postulated as biomarkers for these conditions.

4. Material and methods

4.1. Bacteria Strains

S. aureus NCTC 12493 (MRSA), *S. aureus* ATCC 29213 (MSSA), *S. aureus* ATCC 43300 (MRSA), *S. aureus* ATCC BAA-1707 (MRSA), *S. aureus* ATCC BAA-1717 (MRSA), *S. aureus* ATCC BAA-1747 (MRSA), *S. aureus* ATCC BAA-1761 (MRSA), *S. aureus* ATCC BAA-1766 (MRSA), *S. aureus* ATCC BAA-1768 (MRSA), *Staphylococcus epidermidis* ATCC 35984, *Escherichia coli* ATCC 25922, *Klebsiella pneumoniae* ATCC 13883, *Proteus mirabilis* ATCC 25933, *Salmonella enterica* ATCC 14028 and *Pseudomona aeruginosa* ATCC 10145 strains were used in this study. 15 clinical isolates belonging to different sequence types (STs) of methicillin-resistant *S. aureus* (ST1 = US400, ST5 = US100/EMRSA-3, ST8 = US300/EMRSA-2, ST30 = US200/EMRSA-16, ST22 = EMRSA-15) were collected by Karolinska Institute (Sweden). Bacteria were initially inoculated onto tryptone soy agar with 5% defibrinated sheep blood (Thermo Scientific, Inc., Waltham, MA, USA) and incubated at 37 °C for 24 h. Then, single colonies of each strain were transferred to Tryptone Soy Broth (Thermo Scientific) and cation-adjusted Mueller Hinton Broth (Thermo Scientific) for the nuclease activity assay or antimicrobial susceptibility testing, respectively.

4.2. Synthesis of floxuridine phosphoramidite

Floxuridine was purchased from Interchim, France. Stocks solutions of floxuridine at a concentration of 1 mg/mL were prepared according to manufacturer's recommendations. The synthesis of floxuridine phosphoramidite is schematically described in Fig. S2. Floxuridine (1) was selectively protected with DMTrCl at the 5′ position, yielding 73% product (2) followed by the addition of 2-cyanoethyl-N,N-diiso-propyl-chlorophosphoramidite in 1,8-diazabicyclo[5.4.0]undec-7-ene (DBU), with a yield of 62% (3).

4.3. Oligonucleotide synthesis

All oligonucleotides were synthesized by the standard method of solid-phase phosphoramidite chemistry, followed by high-performance liquid chromatography (HPLC) purification. Next, successful oligonucleotide synthesis was confirmed by matrix-assisted laser desorption ionization mass spectrometry (MALDI-MS). The purity of the oligonucleotides is typically >95%. The list of the oligonucleotides used in this study is provided in Table S1.

4.4. Calculation of equivalent concentration of NA and TOUCANs

Equal amounts in moles for nucleoside analogues (NA) and TOUCANs were tested. The stock solutions for floxuridine were prepared to a final concentration of 1 mg/mL. The stock solutions of TOUCANs were calculated as described below: For each TOUCAN we first calculated their concentration factor (*CF*):

CF = Molecular Weight (MW) TOUCAN / MW NA.

7mFx *CF* = 2134 g/mol / 246,19 g/mol = 8,67.

Next, TOUCANs with equivalent amount of NA (E) were prepared as follows:

E = NA stock solution x *CF*.

7mFx E = 1 mg/mL x 8,67 = 8,67 mg/mL, containing 1 mg/mL of floxuridine.

In this study, to facilitate the comparison and dose preparations of NAs and TOUCANs at equivalent concentrations, we have used 1 mg/mL containing NA of each TOUCAN, along with 1 mg/mL stock concentration of NAs.

4.5. Nuclease activity assay

Nuclease activity of different *S. aureus* strains was evaluated using the TTprobe and TTprobe2, along with the *S. epidermidis* as negative control and micrococcal nuclease (MN) as positive control. Single colonies of each bacteria strain isolated on tryptone soy agar were transferred to 50 mL Tryptone Soy Broth and incubated at 37 °C for 24 h with 200 rpm shaking. After the incubation period, 100 μL volumes of liquid cultures were transferred to fresh 50 mL Tryptone Soy Broth (1/500 dilution) and incubated under the same conditions for 12 h. Next, cultures were centrifuged at 4500*g* for 30 min and the supernatants were used for the nuclease activity assay. Briefly, 8 μL of supernatants were mixed with 1 μL Phosphate Buffer Solution supplemented with calcium and magnesium (PBS +/+) and 1 μL TTprobe or TTprobe2. The reactions were incubated at 37 °C for 1 h. Then, 290 μL of 10 mM EDTA in PBS without calcium and magnesium (PBS−/−) solution were added to stop the reaction. The final solution was loaded in triplicate wells (95 μL each) in a 96-well black flat-bottom polystyrene plate (Thermo Scientific). Fluorescence intensity was acquired at 485/20 excitation and 528/20 emission filter pair using Cytation 1 Cell Imaging Multimode Reader (Biotek Instruments Inc., USA).

4.6. Antimicrobial susceptibility testing (AST)

Minimal inhibitory concentrations (MIC) of floxuridine and TOUCAN 7mFx against each bacteria strain were determined using broth microdilution method. A single colony obtained from overnight culture on tryptone soy agar with 5% defibrinated sheep blood was inoculated into cation-adjusted Mueller Hinton Broth and adjusted to 10^6 CFU/mL. TOUCANs were prepared in PBS+/+ (treatment) and serial dilutions were made in 96-well polystyrene round bottom microwell plates (Thermo Scientific) with 50 μL final volume for each concentration. PBS+/+ was used as untreated control. Next, 50 μL bacteria solutions were added to each well to a final bacteria concentration of 5×10^5 CFU/mL. Plates were incubated at 37 °C in the Cytation 1 Cell Imaging Multimode Reader. Growth curves were obtained by measuring optical densities at 600 nm every 10 min for 18 h using Cytation 1 Cell Imaging Multimode Reader. In addition, visual inspection of the plates was carried out after 18 h and photographs were taken. MICs were defined as the lowest drug concentration in the microwell with no visible bacterial growth at the end of the incubation period.

4.7. Mouse serum preparation

Blood was collected from C57/BL6 mice by cardiac puncture prior to euthanasia. Next, to prevent haemolysis, the needle was removed from the syringe and the blood was slowly transferred to an Eppendorf tube. Approximately 500 μL blood was obtained from each mouse. Blood samples were allowed to clot for 2 h at room temperature, followed by centrifugation at 3.000 rpm for 15 min. The upper layer of serum was

carefully transferred into a fresh 1.5 mL Eppendorf tube and stored at −20 °C for later analysis.

4.8. Mass spectrometry (MALDI-TOF-MS)

The nuclease activity assay for mass spectrometry analysis was optimized for *S. aureus* nuclease (MN) and mouse serum, as follows: i) *S. aureus*, 1 μL of MN (1 U/μL) in 7 μL of ultra-pure water (Chem-Lab) was incubated from 0 to 40 min at room temperature with 2 μL of TOUCAN 7mFx (at a concentration of 20 mg/L). A sample without MN was used as control. ii) Mouse serum samples were prepared by adding 1 μL of serum to 7 μL of ultra-pure water, followed by adding 2 μL of TOUCAN 7mFx (at a concentration of 20 mg/L) and incubated for 0 to 40 min at room temperature. Next, 0.5 μL of each sample was loaded directly onto a Standard μFocus MALDI plate (Hudson Surface Technology), and supplemented with 0.5 μL of matrix solution. The matrix solution was prepared in two steps: i) a solution of 5 mg of citric acid ammonium salt in 1 mL of ultra-pure water was added to a 10 mg of 2,4,6-Trihydroxyacetophenone (THAP). ii) the previous solution was dissolved 1:1 with acetonitrile. MALDI-TOF-MS analysis was performed using an UltrafleXtreme III time-of-flight mass spectrometer equipped with a Nd:YAG laser (Smartbeam II, 355 nm, 1 kHz) and controlled by Flex Control 3.3 software (Bruker Daltonics, Bremen, Germany). The acquisitions were carried out in negative-ion linear mode at a laser frequency of 1 kHz. All the spectra were analyzed using FlexAnalysis software 3.0 (Bruker Daltonics, Bremen, Germany).

4.9. Mouse acute tolerability studies

A total of eight female C57/BL6 mice, at the age of six weeks, were housed in pairs per cage. The mice were administered with intravenous treatment of either floxuridine or TOUCAN 7mFx drugs in four doses given every six hours. Two different concentrations of the drugs, 5 mg/kg or 25 mg/kg, were tested, resulting in a total daily dose of 20 mg/kg or 100 mg/kg, respectively. The mice were assessed using a clinical scoring system based on their observed behaviour and physical manifestations, with scores ranging from 0 to 4, as previously described [56]. A score of 0 indicated the mice were in good health, whereas scores 1 to 3 represented varying degrees of clinical signs, such as reduced movements, piloerection, and pinched eyes. A score of 4 indicated severe distress, which necessitated sacrificing the mice (See Table S2 for detailed descriptions).

4.10. In vivo assessment of the therapeutic efficacy of TOUCANs in a mouse infection model

All animal experiments described in this study were conducted at the Linköping University Centre for Biomedical Resources and were approved by the Swedish Board of Agriculture under the animal protocol No. 1253. All experiments were conducted according to the principles expressed in the Declaration of Helsinki. Six-weeks old female C57/BL6 mice were divided into three different groups: placebo (PBS solution), floxuridine alone and TOUCAN 7mFx, with 8 mice per group. Infection was induced with diluted broth cultures of *S. aureus*. First, *S. aureus* ATCC BAA-1766 strain was cultured on tryptone soy agar with 5% defibrinated sheep blood, single colonies were transferred to 25 mL Tryptone Soy Broth and incubated at 37 °C for 12 h with 200 rpm shaking. Next, bacteria solution was diluted 1/100 using PBS+/+ ($\sim 10^7$ CFU/mL) and 50 μL ($\sim 5 \times 10^5$ CFU) were injected into the right thigh muscles of all mice under sterile conditions and 2% isoflurane anaesthesia. This method was optimized based on our previous animal infection model (9). In this model, the left uninfected leg was used as control for each animal. This model induces reproducible infections after 24 h. Next, the therapeutic regime was adapted from a previous study that validated the efficacy of several antibiotics against *S. aureus* (35). We have treated the mice for 24 h with four doses of TOUCAN 7mFx equivalent to 5 mg/kg of floxuridine in 100 μL of PBS, administered intravenously by tail vein injections. The first dose was administered at 2 h post-inoculation, followed by 3 additional injections at the same concentration every 6 h. In total, each animal was given TOUCAN 7mFx equivalent to 20 mg/Kg/24 h of floxuridine. In this study, two additional groups, placebo and floxuridine alone, were also tested in infected animals. In the placebo group, mice were injected with PBS and in the floxuridine group, mice were injected with 20 mg/Kg/24 h of floxuridine under the same conditions indicated for TOUCAN 7mFx. After 24 h, blood samples were collected, and mice were euthanized. For a thorough blood analysis, we included an additional group of uninfected/untreated mice as control. Next, gross pictures of both legs were taken and samples for histopathological and bacteriological analyses were collected. At least four mice per group per analysis were used in this study.

4.11. Complete blood counts

The mice were anesthetized by an intraperitoneal injection with a combination of 75 mg/Kg ketamine and 1 mg/Kg of dexmedetomidine. The efficacy of the anaesthesia was confirmed by the loss of righting reflex of the animals. Next, mice were dorsally placed on the surgery pad and a 29-gauge needle fitted to a 1 mL syringe was slowly inserted below the sternum. Once the needle reached the mouse heart and blood streamed in the syringe, at least 400 μL of blood were extracted. Next, mice were euthanized. The needle was removed from the syringe and the blood was transferred to vials with EDTA to perform complete blood analysis using the Exigo BM800 veterinary hematology analyzer (Boule, Sweden).

4.12. Gross pictures

Once the mice were euthanized, the skin from legs and dorsal part was removed very carefully to permit observation of the bacterial infection and the associated tissue damage. Next, photographs were taken to document the localized leg infection in each animal.

4.13. Histopathological analysis

To evaluate the histopathological features of infected and uninfected tissues, the thigh muscles of the animals were dissected. Subsequently, tissues were transferred to 4% buffered formaldehyde solution and stored at 4 °C for 24 h. Then, tissues were removed from the formaldehyde solution and transferred into 50% ethanol for 15 min followed by 70% ethanol, until processed. Next, tissues were fixed, dehydrated and wax infiltrated using a Leica TP1020 system for 7 h. Subsequently, tissues were embedded in paraffin using a Leica EG 1150 system. The sections were prepared using an Automatic Rotary Microtome HM355S system and finally, samples were stained with Hematoxylin-Eosin (H&E) using an Automatic Leica ST4020 Linear Stainer. The slides were imaged using an Olympus BX51 light microscope.

4.14. Bacteria counts

To measure viable bacteria count, infected and uninfected thigh muscle tissues from both legs were collected, transferred to individual 1.5 mL vials and weighted on a microbalance. Each sample was homogenized in 1 mL of PBS using Fisherbrand Bead Mill 4 automatic homogenizer (Fisher Scientific). The homogenized solutions were serially diluted to 1:100, 1:1.000, 1:10.000 and 1:100.000. Next, 10 μL of each dilution were inoculated onto three different blood agar plates (tryptone soy agar with 5% defibrinated sheep blood) and incubated for 24 h at 37 °C. After incubation, the bacteria colonies were counted using aCOLyte 3 HD automatic colony counter (Synbiosis) and were reported as Colony Forming Units per gram tissue (CFU/g).

4.15. Laboratory evolution of *S. aureus* under exposure with either floxuridine or TOUCAN 7mFx

We have used broth microdilution (BMD) test to evaluate the evolution of resistance under laboratory conditions, based on the protocol described by Maeda et al. [52]. To begin, the starting concentration of *S. aureus* was determined by measuring the optical density (OD) at 600 nm. A value of 1 OD of McFarland turbidity (~3×10^8 CFU/ml) was used to calculate the desired bacterial concentration, which was achieved by dilution to 1×10^6 CFU/ml. MIC was determined using the same experimental conditions as described in the AST section of this study. The dilution coefficient was calculated by dividing the OD of the micro-well with the highest drug concentration, in which bacterial growth was observed at 1/2 MIC, by the calculated OD of the starting bacterial concentration (0.00045). Bacteria from this well were used as inoculum for the next day of BMD testing. The experiment was carried out for 6 consecutive days under exposure with either floxuridine or TOUCAN 7mFx. All BMD tests were performed in triplicates, and the results were recorded as the MIC value. Resistance evolution was evaluated based on fold MIC shift over the consecutive days.

4.16. Statistics

Statistical analysis was performed in GraphPad Prism V8 (GraphPad). All data were reported as mean ± SD. Differences between groups were assessed by the one-way ANOVA as indicated in the figures using Dunnett's multiple comparison test. $P < 0.05$ was considered as the statistical significance level.

SI contains supporting tables and figures related to the list of oligonucleotides used in this study, MIC data from several ATCC *S. aureus* strains and other ATCC non-targeted bacteria, Blood counts, TOUCAN structure and synthesis, nuclease activity profile of *S. aureus* strains, raw data of TOUCAN response and MICs, stability studies, therapeutic efficacy of TOUCANs, histopathology data, bacterial counts, and acute toxicity (PDF). Animation of TOUCAN's mechanism of action (Video) Supplementary data to this article can be found online at https://doi.org/10.1016/j.jconrel.2023.07.057.

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CRediT authorship contribution statement

Baris A. Borsa: Visualization, Methodology, Formal analysis, Investigation, Writing - review & editing. **Luiza I. Hernandez:** Writing – original draft, Visualization, Methodology. **Tania Jimenez:** Writing – review & editing, Investigation, Formal analysis. **Chaitanya Tellapragada:** Writing – review & editing, Validation, Investigation, Formal analysis. **Christian G. Giske:** Writing – review & editing, Validation, Supervision, Resources, Formal analysis. **Frank J. Hernandez:** Writing – original draft, Supervision, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of Competing Interest

BAB, LIH, FJH are inventors in patent that describes the use of TOUCAN technology andfounders of Theralia AB, a spin-off company from LiU that develops TOUCANs for clinical applications. The other authors declare no competing financial interests.

Data availability

Data will be made available on request.

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